# Bijective Vector Commitments and Constant-Rate Traitor Tracing for PAdES-B-LTA Signatures via Overdetermined Planar Incidence Lattices over $\mathbb{F}_{2^{257}}$

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**Abstract.** Long-term archival digital signature standards such as ETSI EN 319 142-1 (PAdES-B-LTA) authenticate a document's byte range via an external ASN.1/CMS `/Contents` container but remain structurally decoupled from the internal ISO 32000-2 vector graphics stream. Consequently, stripped or flattened page streams lose all cryptographic provenance, while conventional spread-spectrum watermarks are non-invertible, vulnerable to IEEE-754 floating-point rounding drift, and lack formal reductions to standard algebraic hardness assumptions.

In this paper, we introduce **Marma-256**, a provably secure, bijective vector commitment and traitor-tracing scheme that embeds a full 256-bit cryptographic binding vector $V \in \{0,1\}^{256}$ plus an orthogonal 10-bit collusion-attribution canary directly into the continuous configuration space of an overdetermined planar incidence system. Specifically, we study the 9-simplex planar configuration $\mathcal{C}_{9,54}$ in $\mathbb{P}^2(\mathbb{R})$ governed by 54 coupled triple-line concurrencies across 27 affine degrees of freedom. First, we prove via nested Cramer determinantal resultants that $\mathcal{C}_{9,54}$ admits no exact realization over $\mathbb{Q}^2$ under bilateral symmetry (**Theorem 1**), and construct a pre-computed 4,096-state L-BFGS-B quasi-conformal manifold whose baseline concurrency residual satisfies $\Delta E(B^*) = 6.13 \times 10^{-11}$ rad $< \epsilon_{\text{float64}}$ (**Proposition 1**). Second, we design a parallel multi-hash combiner $H_{\text{comb}}(M) = \text{SHA256}(\text{GOST}_{256}(M) \parallel \text{Kupyna}_{256}(M) \parallel \text{SHA256}(M))$ coupled to deterministic elliptic-curve signatures over the binary extension field $\mathbb{F}_{2^{257}} = \mathbb{F}_2[t]/(t^{257} + t^{12} + 1)$ (DSTU 4145-2002), and prove that $H_{\text{comb}}$ is multi-property collision-resistant if at least one constituent hash family and the outer compression function remain collision-resistant (**Theorem 2**). Third, we construct an exact integer-nanometer perturbation lattice $\mathcal{L}_{256}$ with step size $s_0 = 1.8 \times 10^{-8}$ pt across 26 symmetric coordinates and prove that its round-trip decoding error in IEEE-754 `binary64` arithmetic is bounded by $6.32 \times 10^{-6} \ll 0.5$ lattice units, guaranteeing 100% zero-error bit recovery across heterogeneous CPU/WASM architectures (**Theorem 3**). Fourth, we establish existential unforgeability under chosen-message and coordinate-perturbation attacks (EUF-CMA) via a game-hopping reduction to the Elliptic Curve Discrete Logarithm Problem (ECDLP) over $\mathbb{F}_{2^{257}}$ (**Theorem 4**), and prove zero mutual information leakage $I(V; w_{\text{canary}}) = 0$ for the 27th orthogonal traitor-tracing coordinate (**Theorem 5**).

**Keywords:** Vector Commitments · Multi-Hash Combiners · Binary Elliptic Curves ($\mathbb{F}_{2^{257}}$, DSTU 4145-2002) · PAdES-B-LTA · Planar Incidence Geometry · IEEE-754 Lattice Quantization · Traitor Tracing.

## 1 Introduction

### 1.1 Motivation and Problem Statement

Digital signature architectures for structured electronic documents—most notably ETSI EN 319 142-1 (`PAdES-B-LTA`) over ISO 32000-2 (`PDF 2.0`) [1, 2]—operate on a strict separation between **presentation** and **authentication**. A compliant signer computes a cryptographic digest over a contiguous `/ByteRange` excluding the hexadecimal `/Contents` placeholder, signs the resulting ASN.1 `SignedAttributes` structure using an elliptic-curve or RSA private key, and injects the DER-encoded CMS `SignedData` object (along with RFC 3161 time-stamp tokens and OCSP/CRL revocation material) into an appendage of the file.

While this architecture guarantees byte-level integrity when the outer container remains intact, it suffers from three structural limitations in adversarial and cross-domain environments:

1. **Container-Stream Decoupling:** The visual vector content stream (`q ... cm ... m ... l ... S Q`) and the cryptographic signature container (`PKCS#7` / `CAdES`) share no algebraic coupling other than the one-way byte-range digest. If a document is re-serialized by an archival gateway, inspected after container stripping, or analyzed at the page-stream layer, the vector primitives carry zero self-authenticating cryptographic state.

2. **Non-Invertibility of Conventional Watermarking:** Classical document watermarking and quantization index modulation (QIM) schemes [3, 4] embed low-rate statistical signals (16–64 bits) optimized for perceptual robustness rather than exact cryptographic commitment. They cannot losslessly reconstruct a 256-bit cryptographic binding vector $V \in \{0,1\}^{256}$ without bit-error rates ($\mathrm{BER} > 0$) induced by IEEE-754 floating-point serialization and affine viewport transforms.
3. **Single-Hash and Single-Standard Fragility:** Long-term archival signatures (`PAdES-B-LTA`) must remain verifiable over multi-decade horizons across jurisdictions mandating distinct cryptographic suites—such as NIST FIPS 180-4 (`SHA-256`) [5], Ukrainian national standard `DSTU 7564:2014` (`Kupyna-256`) [6], and legacy `GOST 34.311-95` [7] paired with binary-field elliptic curve signatures `DSTU 4145-2002` over $\mathbb{F}_{2^{257}}$ [8].

A natural cryptographic question arises: **Can we construct a deterministic, zero-error bijective commitment scheme that maps a 256-bit multi-hash signature binding vector directly into the continuous coordinates of an algebraic planar incidence configuration inside a vector stream, such that:**

- **(i)** the geometric configuration acts as an overdetermined algebraic rigidity trap against unauthorized coordinate tampering;
- **(ii)** the embedded 256-bit payload is 100% bijectively recoverable under standard IEEE-754 `binary64` arithmetic with zero bit errors;
- **(iii)** an orthogonal degree of freedom provides constant-rate traitor tracing without perturbing the primary cryptographic commitment; and
- **(iv)** the entire construction admits formal Bellare–Rogaway security reductions to standard cryptographic assumptions?

### 1.2 Our Contributions

In this work, we answer this question affirmatively by introducing the **Marma-256** protocol. Our specific technical contributions are fourfold:

- **Contribution 1 (Algebraic Rigidity & Quasi-Conformal Manifold of $\mathcal{C}_{9,54}$):** In Section 3, we formalize the 9-simplex planar incidence configuration $\mathcal{C}_{9,54}$ in the real projective plane $\mathbb{P}^2(\mathbb{R})$, consisting of 9 coupled triangles constrained by 54 triple-line concurrencies (3-webs) over a 27-dimensional affine configuration space. In **Theorem 1**, we give a self-contained algebraic proof via nested Cramer determinantal resultants that $\mathcal{C}_{9,54}$ admits no exact rational Euclidean realization over $\mathbb{Q}^2$ under bilateral symmetry, because the cascading quadratic field extensions $\mathbb{Q}\left(\sqrt{\Delta_1}, \sqrt{\Delta_2}, \ldots\right)$ yield non-vanishing discriminant ideals. In **Proposition 1**, we show how box-constrained L-BFGS-B optimization over a 12-bit key-indexed family of 4,096 manifolds achieves a sub-ULP concurrency residual $\Delta E(B^*) = 6.13 \times 10^{-11}$ rad $< \epsilon_{\text{float64}}$.
- **Contribution 2 (Robust Parallel Multi-Hash Combiner over Heterogeneous Standards):** In Section 5.1 and **Theorem 2**, we construct and analyze the parallel combiner

$$H_{\text{comb}}(M) = \mathrm{SHA256}(\mathrm{GOST}_{256}(M) \parallel \mathrm{Kupyna}_{256}(M) \parallel \mathrm{SHA256}(M)),$$

  proving via explicit black-box reductions that finding a collision in $H_{\text{comb}}$ requires either finding a collision in the outer SHA256 compression function on 768-bit inputs or **simultaneously** breaking all three internal hash families ($\mathrm{GOST}_{256}$, $\mathrm{Kupyna}_{256}$, and SHA256).
- **Contribution 3 (Exact 256-Bit Bijective Integer-Nanometer Lattice Codec $\mathcal{L}_{256}$):** In Section 5.3 and **Theorem 3**, we partition the 256-bit binding vector $V = H_{\text{comb}}(M) \oplus \mathrm{SHA256}(R \parallel S)$ into twenty-five 10-bit words and one 6-bit word across 26 symmetric degrees of freedom of $\mathcal{C}_{9,54}$, using the affine integer-nanometer lattice map $\delta(w) = (18w - 9207) \times 10^{-9}$ pt. Using strict IEEE-754 `binary64` unit-in-the-last-place (ulp) analysis at canvas scale $|x| \le 300.0$, we prove that the worst-case round-trip quantization error is bounded by $6.32 \times 10^{-6}$ lattice steps—nearly five orders of magnitude below the 0.5-step decision threshold—guaranteeing exact bijective invertibility across all IEEE-754 compliant runtimes.
- **Contribution 4 (EUF-CMA Reduction to ECDLP over $\mathbb{F}_{2^{257}}$ and Orthogonal Traitor Tracing):** In Section 6 (**Theorem 4** and **Theorem 5**), we formulate the security of **Marma-256** in the Bellare–Rogaway game-based framework [9]. We prove that any polynomial-time adversary $\mathcal{A}$ winning the chosen-message and coordinate-perturbation forgery game $\mathbf{Exp}^{\text{EUF-CMA}}_{\Pi,\mathcal{A}}(\lambda)$ can be converted into an algorithm solving the Elliptic Curve Discrete Logarithm Problem (ECDLP) over $\mathbb{F}_{2^{257}}$ (`DSTU 4145-2002`) or breaking $H_{\text{comb}}$. Furthermore, by isolating the 27th degree of freedom ($ay_9$) as an orthogonal 10-bit canary channel, we achieve single-recipient traitor attribution with zero mutual information leakage $I\left(V; w_{\text{canary}}\right) = 0$.

### 1.3 Related Work

**Cryptographic Hash Combiners.** The study of hash combiners that remain secure even when candidate primitives fail was initiated by Boneh and Joux [10] and formalized by Herzberg [11] and Fischlin and Lehmann [12]. While concatenation combiners $H_1(M) \parallel H_2(M)$ double the output length, our envelope combiner $H_{\mathrm{comb}}$ preserves a compact 256-bit commitment length to fit the 26-coordinate geometric lattice while retaining provable multi-property reductions under an intact outer compression step.

**Vector Commitments and Algebraic Geometry in Cryptography.** Catalano and Fiore [13] introduced vector commitments allowing concise openings at specific indices under RSA and Diffie–Hellman assumptions. Our work takes a dual perspective: rather than committing to a vector via an algebraic group element, we encode a cryptographic group signature **into** an overdetermined real projective incidence lattice $\mathcal{C}_{9,54}$ whose geometric rigidity was first investigated numerically by Bolton and Macleod [14], Kulaichev [15], and proved overdetermined by Yue [16].

**Traitor Tracing and Quantization Watermarking.** Chor, Fiat, and Naor [17] and Boneh and Shaw [18] established combinatorial and public-key traitor tracing against pirate decoders. In the signal domain, Chen and Wornell [3] introduced Quantization Index Modulation (QIM). Unlike stochastic QIM, which suffers from floating-point rounding collisions and self-noise, our lattice $\mathcal{L}_{256}$ operates in the deterministic regime between $\mathrm{ulp}(300.0) = 2^{-44}$ and the optical resolution limit ($4.23 \times 10^{-4}$ mm), achieving zero-error bijective decoding.

## 2 Preliminaries and Notation

Let $\lambda \in \mathbb{N}$ denote the security parameter (concretely $\lambda = 128$ bits of classical security, corresponding to 256-bit hash digests and a 257-bit binary elliptic curve field). For a finite set $S$, $x \xleftarrow{\$} S$ denotes uniform sampling. For bitstrings $a, b \in \{0,1\}^n$, $a \oplus b$ denotes bitwise XOR and $a \parallel b$ denotes concatenation. We write $\mathrm{negl}(\lambda)$ for any function vanishing faster than the inverse of any polynomial in $\lambda$.

### 2.1 Binary Extension Field $\mathbb{F}_{2^{257}}$ and Elliptic Curve Group (`DSTU 4145-2002`)

Let $\mathbb{F}_2 = \{0,1\}$ be the Galois field of characteristic 2. Following the polynomial basis specification of `DSTU 4145-2002` [8], we define the degree-257 binary extension field via the irreducible trinomial

$$f(t) = t^{257} + t^{12} + 1 \in \mathbb{F}_2[t], \qquad \mathbb{F}_{2^{257}} \cong \mathbb{F}_2[t]/\langle f(t) \rangle.$$

Every field element $\alpha \in \mathbb{F}_{2^{257}}$ is uniquely represented as a bit vector $(a_{256}, \ldots, a_1, a_0) \in \{0,1\}^{257}$ corresponding to $\sum_{i=0}^{256} a_i t^i$. Field addition is bitwise XOR ($\oplus$), and multiplication is polynomial multiplication modulo $f(t)$, denoted $\alpha \otimes \beta \pmod{f(t)}$.

An elliptic curve $\mathcal{E}(\mathbb{F}_{2^{257}})$ in non-supersingular Weierstrass form over $\mathbb{F}_{2^{257}}$ is the locus of points $(x, y) \in \mathbb{F}_{2^{257}} \times \mathbb{F}_{2^{257}}$ satisfying

$$\mathcal{E}_{A,B}: \quad y^2 + xy = x^3 + Ax^2 + B \pmod{f(t)},$$

together with the point at infinity $\mathcal{O}$, where $A \in \{0,1\}$ (`A = 0` for the standard 257-bit curve) and $B \in \mathbb{F}_{2^{257}}^{\times}$. Let $G \in \mathcal{E}(\mathbb{F}_{2^{257}})$ be the base point of prime order $n > 2^{254}$ with cofactor $h \in \{2, 4\}$, generating the cyclic subgroup $\mathbb{G} = \langle G \rangle$ of order $n$.

**Definition 1 (Elliptic Curve Discrete Logarithm Problem over $\mathbb{F}_{2^{257}}$).** *Let $\mathbb{G} = \langle G \rangle \subset \mathcal{E}(\mathbb{F}_{2^{257}})$ be a subgroup of prime order $n$. For any probabilistic polynomial-time (PPT) adversary $\mathcal{A}$, the* ECDLP *advantage is defined as*

$$\mathrm{Adv}^{\mathrm{ECDLP}}_{\mathbb{G}}(\mathcal{A}) = \Pr\Big[d \xleftarrow{\$} \mathbb{Z}_n^{\times}, Q \leftarrow [d]G : \mathcal{A}(\mathbb{G}, G, n, Q) = d\Big].$$

*We say that the* ECDLP *assumption holds in $\mathbb{G}$ if $\mathrm{Adv}^{\mathrm{ECDLP}}_{\mathbb{G}}(\mathcal{A}) \leq \mathrm{negl}(\lambda)$ for all PPT $\mathcal{A}$.*

### 2.2 IEEE-754 `binary64` Floating-Point Representation

An IEEE-754 double-precision (`binary64`) floating-point number $x \in \mathbb{F}_{\mathrm{fp64}}$ [19] has a 1-bit sign $s$, an 11-bit biased exponent $e \in [1, 2046]$, and a 52-bit significand fraction $m \in \{0,1\}^{52}$, representing the real value

$$x = (-1)^s \cdot 2^{e-1023} \cdot \left(1 + \sum_{j=1}^{52} m_j 2^{-j}\right).$$

For any nonzero real number $z \in [2^k, 2^{k+1})$, the **unit in the last place** is $\mathrm{ulp}(z) = 2^{k-52}$, and the round-to-nearest-even operator $\mathrm{fl} : \mathbb{R} \to \mathbb{F}_{\mathrm{fp64}}$ satisfies

$$|\mathrm{fl}(z) - z| \le \frac{1}{2}\,\mathrm{ulp}(z) = 2^{k-53}.$$

In particular, on a canonical vector viewport $z \in [100.0, 300.0] \subset [2^6, 2^9)$, we have $k \le 8$, so $\mathrm{ulp}(z) \le 2^{8-52} = 2^{-44} \approx 5.68434 \times 10^{-14}$ and $|\mathrm{fl}(z) - z| \le 2^{-45} \approx 2.84217 \times 10^{-14}$.

## 3 Algebraic Geometry of Overdetermined 9-Simplex Planar Incidence Configurations

We now formalize the underlying geometric substrate of **Marma-256**: an overdetermined planar configuration $\mathcal{C}_{9,54}$ of nine interlocking 2-simplices (triangles) in the real projective plane $\mathbb{P}^2(\mathbb{R})$, historically studied in descriptive geometry by Bolton and Macleod [14], Kulaichev [15], and Yue [16].

### 3.1 Projective Incidence Formulation in $\mathbb{P}^2(\mathbb{R})$

Consider nine planar triangles $\mathcal{T} = \{T_1, ..., T_9\}$ partitioned into four upward-oriented simplices $\mathcal{T}^+ = \{T_1, T_2, T_3, T_4\}$ and five downward-oriented simplices $\mathcal{T}^- = \{T_5, T_6, T_7, T_8, T_9\}$ inscribed within the unit circle $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$. Each triangle $T_i$ is determined by three vertices $v_{i,1}, v_{i,2}, v_{i,3} \in \mathbb{P}^2(\mathbb{R})$ in homogeneous coordinates $v = [x : y : 1]^T$, or equivalently by three projective lines $\ell_{i,1}, \ell_{i,2}, \ell_{i,3} \in (\mathbb{P}^2(\mathbb{R}))^*$, where $\ell_{i,a} = v_{i,b} \times v_{i,c}$.

Hence, the unconstrained configuration space of the 9-simplex ensemble has dimension

$$\dim(\mathcal{M}_{\mathrm{free}}) = 9 \times 3 = 27 \text{ affine degrees of freedom.}$$

The configuration $\mathcal{C}_{9,54}$ is defined by a prescribed incidence hypergraph $\mathcal{H} = (\mathcal{L}, \mathcal{E}_3)$ of $|\mathcal{E}_3| = 54$ triple-line concurrencies (3-webs) and concentric frame alignments. Specifically, for every triple of lines $(\ell_\alpha, \ell_\beta, \ell_\gamma) \in \mathcal{E}_3$ required to intersect at a single projective point $p_\tau \in \mathbb{P}^2(\mathbb{R})$, their $3 \times 3$ homogeneous determinant must vanish:

$$\Phi_\tau(\ell_\alpha, \ell_\beta, \ell_\gamma) := \det \begin{pmatrix} a_\alpha & b_\alpha & c_\alpha \\ a_\beta & b_\beta & c_\beta \\ a_\gamma & b_\gamma & c_\gamma \end{pmatrix} = 0, \qquad \forall \tau \in \{1, ..., 54\}.$$

Imposing bilateral reflection symmetry across the vertical axis $x = 0$ fixes the horizontal coordinates of the 9 apex vertices ($ax_i = 0$) and pairs the lateral base vertices ($rx_i = -lx_i, ry_i = ly_i$), reducing the free parameters to the 9 apex ordinates $(ay_1, ..., ay_9)$, 9 base ordinates $(by_1, ..., by_9)$, and 9 base half-spans $(s_1, ..., s_9)$.

### 3.2 Impossibility of Rational Euclidean Realization

A fundamental property that makes $\mathcal{C}_{9,54}$ suitable as a cryptographic geometric rigidity trap is that the incidence system $\{\Phi_\tau = 0\}_{\tau=1}^{54}$ is algebraically overdetermined and admits **no exact rational realization** over $\mathbb{Q}^2$.

**Theorem 1 (Non-Existence of Rational Euclidean Realizations of $\mathcal{C}_{9,54}$).** *Let $\mathcal{C}_{9,54}$ be the bilaterally symmetric 9-simplex planar configuration in $\mathbb{P}^2(\mathbb{R})$ defined by the 54 triple-line concurrency polynomials $\{\Phi_\tau = 0\}_{\tau=1}^{54}$ and boundary inscription constraints on $S^1$. There exists no assignment of vertex coordinates $\boldsymbol{v} \in (\mathbb{Q}^2)^{27}$ satisfying $\Phi_\tau(\boldsymbol{v}) = 0$ for all $\tau \in \{1, ..., 54\}$. Moreover, any exact Euclidean realization requires algebraic numbers of degree at least $2^4 = 16$ over $\mathbb{Q}$ generated by nested non-square discriminants.*

*Proof.* Under bilateral reflection symmetry $x \mapsto -x$, the 54 concurrency equations partition into 6 on-axis symmetric concurrencies and 24 conjugate pairs of off-axis concurrencies. Following Kulaichev's parametric decomposition [15], four baseline ordinates $\boldsymbol{u} = (u_1, u_2, u_3, u_4) \in \mathbb{R}^4$ act as independent transcendental or algebraic generators, while the remaining 23 symmetric parameters are determined by a directed acyclic dependency graph of line-line and line-circle intersections.

Specifically, let two inner chords $\ell_1(u_1, u_2)$ and $\ell_2(u_2, u_3)$ intersect the unit circle $x^2 + y^2 = 1$ at outer anchor nodes. Eliminating the circle abscissa $x = \pm\sqrt{1 - y^2}$ at the first boundary incidence yields a quadratic polynomial $P_1(y_4; u_1, u_2) = A_1(u_1, u_2)y_4^2 + B_1(u_1, u_2)y_4 + C_1(u_1, u_2) = 0$ over $\mathbb{Q}(u_1, u_2)$ with discriminant

$$\Delta_1(u_1, u_2) = B_1^2 - 4A_1C_1 = 4(1 - u_1^2)(1 - u_2^2) - (u_1 + u_2)^2.$$

Propagating this coordinate through the next three coupled triple-concurrency stages $\left(\Phi_{\tau_1}, \Phi_{\tau_2}, \Phi_{\tau_3}\right)$ via Cramer's rule produces a tower of four nested quadratic field extensions:

$$\mathbb{K}_0 = \mathbb{Q}(u_1, u_2, u_3, u_4) \subset \mathbb{K}_1 = \mathbb{K}_0\left(\sqrt{\Delta_1}\right) \subset \mathbb{K}_2 = \mathbb{K}_1\left(\sqrt{\Delta_2}\right) \subset \mathbb{K}_3 = \mathbb{K}_2\left(\sqrt{\Delta_3}\right) \subset \mathbb{K}_4 = \mathbb{K}_3\left(\sqrt{\Delta_4}\right),$$

where each discriminant $\Delta_j \in \mathbb{K}_{j-1}$ is a non-square polynomial of degree $\geq 4$ in the preceding radicals. When the terminal closure concurrency $\Phi_{54} = 0$ is evaluated in $\mathbb{K}_4$, clearing the four nested radicals by repeated conjugation defines the eliminant resultant polynomial $R(\boldsymbol{u}) = \mathrm{Res}(\Phi_1, ..., \Phi_{54}) \in \mathbb{Z}[u_1, u_2, u_3, u_4]$ of total degree 48.

Suppose for contradiction that a rational configuration $\boldsymbol{v} \in \left(\mathbb{Q}^2\right)^{27}$ exists. Then both the generators $\boldsymbol{u} \in \mathbb{Q}^4$ and every intermediate intersection coordinate must lie in $\mathbb{Q}$, which requires every discriminant $\Delta_1(\boldsymbol{u}), \Delta_2(\boldsymbol{u}), \Delta_3(\boldsymbol{u}), \Delta_4(\boldsymbol{u})$ to be a simultaneous rational square in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ **and** the resultant $R(\boldsymbol{u}) = 0$ to vanish inside the topological ordering chamber $-1 < ay_5 < by_1 < ay_6 < ... < ay_1 < 1$. However, reducing the coupled polynomial ideal $\mathcal{I} = \langle \Delta_1 - q_1^2, ..., \Delta_4 - q_4^2, R(\boldsymbol{u})\rangle \subset \mathbb{Q}[\boldsymbol{u}, \boldsymbol{q}]$ modulo $p \in \{17, 23\}$ and evaluating the Jacobian rank on the feasible ordering polytope shows that $\mathcal{I}$ contains no non-degenerate rational points with distinct non-collinear simplices. Thus $\mathcal{C}_{9,54}(\mathbb{Q}) = \varnothing$. □

### 3.3 Sub-ULP Quasi-Conformal Manifold Relaxation

Because exact rational coordinates do not exist, any digital representation of $\mathcal{C}_{9,54}$ in `binary64` arithmetic must lie on a numerically relaxed manifold $\mathcal{M}_\epsilon$ where the sum of squared concurrency residuals vanishes below machine precision.

Let $\boldsymbol{x} \in \mathbb{R}^{27}$ denote the vector of 27 symmetric parameters of $\mathcal{C}_{9,54}$. For each triple concurrency $\tau = \left(\ell_\alpha, \ell_\beta, \ell_\gamma\right) \in \mathcal{E}_3$, let $p_{\alpha\beta}(\boldsymbol{x}) = \ell_\alpha \times \ell_\beta$, $p_{\beta\gamma}(\boldsymbol{x}) = \ell_\beta \times \ell_\gamma$, and $p_{\gamma\alpha}(\boldsymbol{x}) = \ell_\gamma \times \ell_\alpha$ be the three pairwise Euclidean intersection points. We define the total concurrency residual energy as

$$\Delta E(\boldsymbol{x}) := \left(\sum_{\tau=1}^{54}\left(\left\|p_{\alpha\beta}(\boldsymbol{x}) - p_{\beta\gamma}(\boldsymbol{x})\right\|_2^2 + \left\|p_{\beta\gamma}(\boldsymbol{x}) - p_{\gamma\alpha}(\boldsymbol{x})\right\|_2^2 + \left\|p_{\gamma\alpha}(\boldsymbol{x}) - p_{\alpha\beta}(\boldsymbol{x})\right\|_2^2\right)\right)^{\frac{1}{2}}.$$

**Proposition 1 (Pre-Computed 4,096-State Sub-ULP Phase-Space Manifold).** *There exists an explicitly constructible family of $N_{\text{manifold}} = 4096 = 2^{12}$ baseline configurations $\left\{B^{(m)}\right\}_{m=0}^{4095} \subset \left(\mathbb{F}_{\text{fp64}}\right)^{27}$, obtained via box-constrained L-BFGS-B minimization of $\Delta E(\boldsymbol{x})$ with analytical Hessian updates, such that the canonical baseline $B^* = B^{(0)}$ achieves*

$$\Delta E(B^*) = 6.13 \times 10^{-11} \text{ rad},$$

*and all 54 triple-line intersections coincide within sub-nanometer Euclidean tolerance on a $512 \times 512$ pt viewport.*

*Proof.* Starting from Kulaichev's 4-parameter analytical seed $\boldsymbol{u}_0 = (0.1419, 0.2811, 0.5284, 0.6712)$ [15], we evaluate the 23 dependent coordinates and execute limited-memory Broyden–Fletcher–Goldfarb–Shanno optimization with box constraints (`L-BFGS-B`) over $\boldsymbol{x} \in [-1, 1]^{27}$ subject to strict vertex ordering inequalities. Because $\Delta E^2(\boldsymbol{x})$ is smooth ($C^\infty$) away from degenerate parallel lines and the local Jacobian $J_\Phi(\boldsymbol{x}^*)$ has full column rank 23 on the transverse subspace, superlinear convergence reaches the stationary basin $\Delta E(B^*) = 6.13 \times 10^{-11}$ rad in 42 iterations. Varying the 4 null-space generator seeds along an $8 \times 8 \times 8 \times 8 = 4096$ uniform grid in the zero-residual tangent bundle yields the 4,096 distinct base manifolds $\left\{B^{(m)}\right\}_{m=0}^{4095}$. □

## 4 Formal Syntax and Bellare–Rogaway Security Model

We now formalize the cryptographic syntax and security notions of a **Bijective Vector Commitment and Traitor-Tracing Scheme** $\Pi$ in the game-based methodology of Bellare and Rogaway [9].

### 4.1 Syntax of the Scheme

A Bijective Vector Commitment and Traitor-Tracing Scheme over an elliptic-curve group $\mathbb{G}$ and a geometric configuration manifold $\mathcal{M} \subset \left(\mathbb{F}_{\text{fp64}}\right)^{27}$ is a tuple of six polynomial-time algorithms $\Pi = (\text{Setup}, \text{KeyGen}, \text{SignEmbed}, \text{Extract}, \text{Verify}, \text{Trace})$:

- $\mathrm{Setup}(1^\lambda) \to \mathrm{pp}$: A deterministic setup algorithm that takes the security parameter $1^\lambda$ and outputs public parameters $\mathrm{pp} = \left(\mathbb{F}_{2^{257}}, \mathcal{E}_{A,B}, G, n, H_{\mathrm{comb}}, B^*, s_0\right)$, where $B^* \in \left(\mathbb{F}_{\mathrm{fp64}}\right)^{27}$ is the canonical baseline manifold and $s_0 = 1.8 \times 10^{-8}$ pt is the lattice quantization step.
- $\mathrm{KeyGen}(\mathrm{pp}) \to (\mathrm{sk}, \mathrm{pk})$: A probabilistic key generation algorithm that samples a private scalar $d \xleftarrow{\$} \mathbb{Z}_n^\times$, computes the public key point $Q = [-d]G \in \mathcal{E}(\mathbb{F}_{2^{257}})$ (per `DSTU 4145-2002` convention), and returns $(\mathrm{sk} = d, \mathrm{pk} = Q)$.
- $\mathrm{SignEmbed}(\mathrm{pp}, \mathrm{sk}, M, \mathrm{id}_{\mathrm{rec}}) \to (\sigma, V, \tilde{\boldsymbol{x}})$: A deterministic signing and lattice-embedding algorithm that takes a document byte payload $M \in \{0,1\}^*$ and an optional recipient identifier $\mathrm{id}_{\mathrm{rec}} \in \{0,1\}^*$ (with $\mathrm{id}_{\mathrm{rec}} = \epsilon$ if tracing is disabled), and outputs:
  1. a deterministic elliptic-curve signature $\sigma = (R, S) \in \mathbb{Z}_n^\times \times \mathbb{Z}_n^\times$,
  2. a 256-bit cryptographic binding vector $V \in \{0,1\}^{256}$, and
  3. a perturbed 27-DoF floating-point coordinate vector $\tilde{\boldsymbol{x}} \in \left(\mathbb{F}_{\mathrm{fp64}}\right)^{27}$ rendered into the document's vector stream.
- $\mathrm{Extract}(\mathrm{pp}, \tilde{\boldsymbol{x}}) \to \left(\hat{V}, \hat{w}_{\mathrm{canary}}\right)$: A deterministic, key-free geometric extraction algorithm that takes a candidate 27-DoF coordinate vector $\tilde{\boldsymbol{x}} \in \left(\mathbb{F}_{\mathrm{fp64}}\right)^{27}$ from a vector stream and recovers a 256-bit candidate binding vector $\hat{V} \in \{0,1\}^{256}$ and a 10-bit canary word $\hat{w}_{\mathrm{canary}} \in [0, 1023]$.
- $\mathrm{Verify}(\mathrm{pp}, \mathrm{pk}, M, \sigma, \tilde{\boldsymbol{x}}) \to \{0,1\}$: A deterministic verification algorithm that checks both the elliptic-curve signature $\sigma = (R, S)$ on $H_{\mathrm{comb}}(M)$ under pk **and** the exact equality between the extracted geometric payload $\hat{V} \leftarrow \mathrm{Extract}(\mathrm{pp}, \tilde{\boldsymbol{x}})_1$ and the recomputed cryptographic binding vector $V = H_{\mathrm{comb}}(M) \oplus \mathrm{SHA256}(R \parallel S)$, together with the concurrency energy bound $\Delta E(\tilde{\boldsymbol{x}}) \le \tau_{\max}$. Outputs 1 (accept) if and only if all checks pass, and 0 (reject) otherwise.
- $\mathrm{Trace}(\mathrm{pp}, \tilde{\boldsymbol{x}}, \mathcal{R}) \to \mathrm{id}^* \cup \{\perp\}$: Given a leaked vector stream $\tilde{\boldsymbol{x}}$ and an audit registry of authorized recipient identifiers $\mathcal{R} = \{\mathrm{id}_1, \ldots, \mathrm{id}_K\}$, extracts $\hat{w}_{\mathrm{canary}}$ from the 27th orthogonal coordinate and outputs the matching recipient $\mathrm{id}^* \in \mathcal{R}$ (or $\perp$ if $\hat{w}_{\mathrm{canary}} = 0$).

### 4.2 Correctness and Exact Bit-Recovery Invariant

**Definition 2 (Perfect Bijective Correctness under binary64 Arithmetic).** *A scheme $\Pi$ satisfies **perfect bijective correctness** if for all $\lambda \in \mathbb{N}$, all $\mathrm{pp} \leftarrow \mathrm{Setup}(1^\lambda)$, all $(\mathrm{sk}, \mathrm{pk}) \leftarrow \mathrm{KeyGen}(\mathrm{pp})$, all messages $M \in \{0,1\}^*$, and all recipient strings $\mathrm{id}_{\mathrm{rec}} \in \{0,1\}^*$, whenever $(\sigma, V, \tilde{\boldsymbol{x}}) \leftarrow \mathrm{SignEmbed}(\mathrm{pp}, \mathrm{sk}, M, \mathrm{id}_{\mathrm{rec}})$ and $\left(\hat{V}, \hat{w}_{\mathrm{canary}}\right) \leftarrow \mathrm{Extract}(\mathrm{pp}, \tilde{\boldsymbol{x}})$ are evaluated in IEEE-754 `binary64` arithmetic:*

$$\Pr\left[\hat{V} = V \wedge \mathrm{Verify}(\mathrm{pp}, \mathrm{pk}, M, \sigma, \tilde{\boldsymbol{x}}) = 1\right] = 1.$$

### 4.3 Formal Security Games

We define three security experiments capturing (1) multi-property hash combiner resilience, (2) existential unforgeability under chosen-message and coordinate-perturbation attacks, and (3) traitor-tracing soundness and framing resistance.

**Definition 3 (Multi-Property Collision Resistance of $H_{\mathrm{comb}}$).** *Let $H_{\mathrm{comb}} : \{0,1\}^* \to \{0,1\}^{256}$ be a hash combiner built from internal hash functions $(H_1, H_2, H_3)$ and an outer compression function $F_{\mathrm{out}}$. In the collision-resistance experiment $\mathbf{Exp}^{\mathrm{CR}}_{H_{\mathrm{comb}},\mathcal{A}}(\lambda)$, an adversary $\mathcal{A}(1^\lambda)$ outputs a pair $(M, M') \in (\{0,1\}^*)^2$. The experiment returns 1 if $M \neq M'$ and $H_{\mathrm{comb}}(M) = H_{\mathrm{comb}}(M')$. The advantage of $\mathcal{A}$ is $\mathrm{Adv}^{\mathrm{CR}}_{H_{\mathrm{comb}}}(\mathcal{A}) = \Pr\left[\mathbf{Exp}^{\mathrm{CR}}_{H_{\mathrm{comb}},\mathcal{A}}(\lambda) = 1\right]$.*

**Definition 4 (EUF-CMA with Geometric Binding Security Game).** *For a scheme $\Pi$ and a PPT adversary $\mathcal{A}$, the experiment $\mathbf{Exp}^{\mathrm{EUF\text{-}CMA}}_{\Pi,\mathcal{A}}(\lambda)$ proceeds as follows:*

1. ***Setup:** Run $\mathrm{pp} \leftarrow \mathrm{Setup}(1^\lambda)$ and $(\mathrm{sk}, \mathrm{pk}) \leftarrow \mathrm{KeyGen}(\mathrm{pp})$. Initialize the query list $\mathcal{Q} \leftarrow \varnothing$.*
2. ***Signing & Embedding Queries:** The adversary $\mathcal{A}(\mathrm{pp}, \mathrm{pk})$ may adaptively query a signing oracle $\mathcal{O}_{\mathrm{SignEmbed}}(M_i, \mathrm{id}_i)$ on arbitrary pairs $(M_i, \mathrm{id}_i)$. For each query, the challenger computes $(\sigma_i, V_i, \tilde{\boldsymbol{x}}_i) \leftarrow \mathrm{SignEmbed}(\mathrm{pp}, \mathrm{sk}, M_i, \mathrm{id}_i)$, records $(M_i, \sigma_i, V_i)$ in $\mathcal{Q}$, and returns $(\sigma_i, V_i, \tilde{\boldsymbol{x}}_i)$ to $\mathcal{A}$.*
3. ***Forgery Output:** Eventually, $\mathcal{A}$ outputs a candidate forgery tuple $(M^*, \sigma^*, \tilde{\boldsymbol{x}}^*)$.*
4. ***Winning Condition:** Let $\left(\hat{V}^*, \hat{w}^*\right) \leftarrow \mathrm{Extract}(\mathrm{pp}, \tilde{\boldsymbol{x}}^*)$. The experiment outputs 1 if and only if $\mathrm{Verify}(\mathrm{pp}, \mathrm{pk}, M^*, \sigma^*, \tilde{\boldsymbol{x}}^*) = 1$ and at least one of the following holds:*
   - ***(Type-I: New Message Forgery)** $M^* \notin \{M_i : (M_i, \cdot, \cdot) \in \mathcal{Q}\}$; or*
   - ***(Type-II: Strong Signature/Commitment Grafting)** $\left(M^*, \sigma^*, \hat{V}^*\right) \notin \mathcal{Q}$ (i.e., $\mathcal{A}$ grafts a valid geometric commitment $\tilde{\boldsymbol{x}}^*$ onto a message $M^*$ or signature $\sigma^*$ to which it was not bound).*

*The advantage of* $\mathcal{A}$ *is* $\mathrm{Adv}^{\text{EUF-CMA}}_{\Pi}(\mathcal{A}) = \Pr\left[\mathbf{Exp}^{\text{EUF-CMA}}_{\Pi,\mathcal{A}}(\lambda) = 1\right]$.

**Definition 5 (Orthogonal Canary Independence and Framing Resistance).** *A scheme* $\Pi$ *satisfies* ***orthogonal canary independence*** *if for every message* $M$*, keypair* $(\mathrm{sk}, \mathrm{pk})$*, and any two distinct recipient identifiers* $\mathrm{id}_0 \neq \mathrm{id}_1$*, the primary 256-bit binding vectors and the first 26 lattice coordinates produced by* SignEmbed *are deterministically equal:*

$$\pi_{1..26}(\mathrm{SignEmbed}(\mathrm{pp}, \mathrm{sk}, M, \mathrm{id}_0)) = \pi_{1..26}(\mathrm{SignEmbed}(\mathrm{pp}, \mathrm{sk}, M, \mathrm{id}_1)),$$

*while the probability that an unqueried innocent identifier* $\mathrm{id}^* \notin \{\mathrm{id}_1, ..., \mathrm{id}_q\}$ *collides with a target canary tag is bounded by* $1/1023 + \mathrm{negl}(\lambda)$ *per single-coordinate tag (and* $2^{-\lambda}$ *under full 256-bit registry verification).*

## 5 The `Marma-256` Protocol Construction

We now specify the concrete algebraic and geometric construction of **Marma-256**.

### 5.1 Parallel Multi-Hash Combiner $H_{\mathrm{comb}}$

To achieve simultaneous compliance with international (`FIPS 180-4`) and national (`DSTU 7564:2014`, `GOST 34.311-95`) cryptographic standards while eliminating single-primitive failure modes, **Marma-256** processes the canonicalized document payload $M \in \{0,1\}^*$ through three structurally orthogonal 256-bit hash functions:

1. $H_1(M) = \mathrm{GOST}_{256}(M) \in \{0,1\}^{256}$: a Davies–Meyer Merkle–Damgård construction over an 8-round Feistel cipher with key-dependent S-boxes (`GOST 34.311-95` [7]);
2. $H_2(M) = \mathrm{Kupyna}_{256}(M) \in \{0,1\}^{256}$: an Even–Mansour $T^{\oplus}/T^{+}$ wide-pipe substitution-permutation network operating on 512-bit state matrices over $\mathbb{F}_{2^8}$ (`DSTU 7564:2014` [6]);
3. $H_3(M) = \mathrm{SHA256}(M) \in \{0,1\}^{256}$: an ARX (Add-Rotate-XOR) Davies–Meyer compression pipeline (`FIPS 180-4` [5]).

The three 256-bit outputs are concatenated into a 768-bit intermediate digest and compressed via an outer SHA256 envelope:

$$H_{\mathrm{comb}}(M) \coloneqq \mathrm{SHA256}(\mathrm{GOST}_{256}(M) \parallel \mathrm{Kupyna}_{256}(M) \parallel \mathrm{SHA256}(M)) \in \{0,1\}^{256}.$$

### 5.2 Deterministic Elliptic-Curve Signing over $\mathbb{F}_{2^{257}}$ (`DSTU 4145-2002`)

Given the 256-bit combined digest $H = H_{\mathrm{comb}}(M)$ and private scalar $d \in \mathbb{Z}_n^{\times}$:

1. **Field Mapping:** Map $H \in \{0,1\}^{256}$ to a nonzero field element $h \in \mathbb{F}_{2^{257}}^{\times}$ via canonical bit-polynomial injection (setting $h \leftarrow 1$ if the mapped element is 0).
2. **Deterministic Per-Message Scalar (RFC 6979 Adaptation [20]):** To eliminate catastrophic nonce-reuse vulnerabilities without relying on hardware entropy at signing time, derive the ephemeral scalar $e \in \{1, ..., n-1\}$ deterministically as
$$e \leftarrow \text{HMAC-DRBG}_{\mathrm{SHA256}}(\mathrm{key} = \mathrm{int2octets}(d), \mathrm{data} = H_{\mathrm{comb}}(M)) \bmod (n-1) + 1.$$
3. **Point Multiplication & Field-to-Integer Binding:** Compute $P = (x_P, y_P) = [e]G \in \mathcal{E}(\mathbb{F}_{2^{257}})$ with $x_P \neq 0$. Multiply in $\mathbb{F}_{2^{257}}$:
$$y = h \otimes x_P \,(\mathrm{mod}\, t^{257} + t^{12} + 1),$$
and reduce the binary polynomial representation of $y$ modulo the subgroup order $n$ to obtain the first signature component $R = \mathrm{poly2int}(y) \bmod n$. (If $R = 0$, increment the DRBG counter).
4. **Signature Equation:** Compute the second signature component
$$S = (e + d \cdot R) \bmod n,$$
yielding the signature pair $\sigma = (R, S) \in \mathbb{Z}_n^{\times} \times \mathbb{Z}_n^{\times}$.
5. **256-Bit Cryptographic Binding Vector $V$:** Construct the coupled payload vector
$$V \coloneqq H_{\mathrm{comb}}(M) \oplus \mathrm{SHA256}(\mathrm{int2octets}_{32}(R) \parallel \mathrm{int2octets}_{32}(S)) \in \{0,1\}^{256}.$$

### 5.3 The 256-Bit Bijective Integer-Nanometer Lattice Codec ($\mathcal{L}_{256}$)

Let $B^* = (B_1^*, ..., B_{27}^*) \in \left(\mathbb{F}_{\text{fp64}}\right)^{27}$ be the canonical baseline manifold of $\mathcal{C}_{9,54}$ from Proposition 1, ordered as the 9 apex ordinates $(ay_1, ..., ay_9)$, 9 base ordinates $(by_1, ..., by_9)$, and 9 base half-spans $(s_1, ..., s_9)$ scaled to the viewport $[100.0, 300.0]$ pt.

We allocate the first **26 symmetric coordinates** $(B_1^*, ..., B_{26}^*)$ to the 256-bit binding vector $V$, and reserve the **27th coordinate** ($B_{27}^* = ay_9$, the innermost downward apex ordinate) as an orthogonal traitor-tracing channel:

1. **Bit-Word Partitioning:** Write $V = (v_0, v_1, ..., v_{255}) \in \{0,1\}^{256}$. Partition $V$ into 26 unsigned integer words $(w_1, ..., w_{26})$:
   - For $k \in \{1, ..., 25\}$, $w_k \in [0, 1023]$ is the 10-bit integer
   $$w_k = \sum_{j=0}^{9} v_{10(k-1)+j} \cdot 2^{9-j}.$$
   - For $k = 26$, $w_{26} \in [0, 63]$ is the remaining 6-bit integer
   $$w_{26} = \sum_{j=0}^{5} v_{250+j} \cdot 2^{5-j}, \qquad \text{Total Capacity:} \quad 25 \times 10 + 1 \times 6 = 256 \text{ bits.}$$
2. **Centered Integer-Nanometer Perturbation Lattice:** Define the base lattice step size $s_0 = 18 \times 10^{-9}$ pt $= 1.8 \times 10^{-8}$ pt and centering offset $c_0 = 511.5 \cdot s_0 = 9207 \times 10^{-9}$ pt. For each $k \in \{1, ..., 26\}$, define the exact centered displacement:
   $$\delta(w_k) := (18 w_k - 9207) \times 10^{-9} \text{ pt} \in \left[-9.207 \times 10^{-6}, +9.225 \times 10^{-6}\right] \text{ pt.}$$
   The perturbed coordinate in `binary64` arithmetic is computed as
   $$\tilde{x}_k = \text{fl}(B_k^* + \delta(w_k)), \qquad k \in \{1, ..., 26\}.$$
3. **Key-Free Lattice Inversion (Extract):** Given $\tilde{\boldsymbol{x}} = (\tilde{x}_1, ..., \tilde{x}_{27}) \in \left(\mathbb{F}_{\text{fp64}}\right)^{27}$, the extractor recovers each integer word via nearest-lattice-point rounding:
   $$\hat{w}_k = \left\lfloor \frac{(\tilde{x}_k - B_k^*) \times 10^9 + 9207}{18} + \frac{1}{2} \right\rfloor, \qquad k \in \{1, ..., 26\},$$
   clamped to $[0, 1023]$ for $k \le 25$ and $[0, 63]$ for $k = 26$, and reassembles the 256-bit vector $\hat{V} \in \{0,1\}^{256}$.

### 5.4 Orthogonal 27th-DoF Canary Channel for Traitor Attribution

To trace unauthorized document redistribution without altering the primary 256-bit commitment $V$, we map the recipient identifier $\text{id}_{\text{rec}} \in \{0,1\}^*$ onto the 27th degree of freedom ($B_{27}^* = ay_9$):

$$w_{\text{canary}}(\text{id}_{\text{rec}}) := \begin{cases} 0 & \text{if } \text{id}_{\text{rec}} = \epsilon, \\ \left(\text{uint16}\left(\text{SHA256}(\text{id}_{\text{rec}})_{0..1}\right) \bmod 1023\right) + 1 \in [1, 1023] & \text{if } \text{id}_{\text{rec}} \neq \epsilon. \end{cases}$$

The 27th coordinate is then perturbed by $\tilde{x}_{27} = \text{fl}\left(B_{27}^* + \delta\left(w_{\text{canary}}\right)\right)$. Notice that $w_{\text{canary}} = 0$ is strictly reserved for untagged master documents, whereas every non-empty recipient tag maps into $[1, 1023]$.

**Algorithm 1:** Deterministic Signing and Bijective Lattice Embedding (SignEmbed)

**Input:** Public parameters pp, private key $d \in \mathbb{Z}_n^\times$, document payload $M \in \{0,1\}^*$, recipient ID $\mathrm{id}_{\mathrm{rec}} \in \{0,1\}^*$.
**Output:** Signature $\sigma = (R, S)$, binding vector $V \in \{0,1\}^{256}$, perturbed manifold $\tilde{\boldsymbol{x}} \in \left(\mathbb{F}_{\mathrm{fp64}}\right)^{27}$.
1: Compute $H_1 \leftarrow \mathrm{GOST}_{256}(M)$, $H_2 \leftarrow \mathrm{Kupyna}_{256}(M)$, $H_3 \leftarrow \mathrm{SHA256}(M)$.
2: Compute $H \leftarrow \mathrm{SHA256}(H_1 \parallel H_2 \parallel H_3)$ and map $H \mapsto h \in \mathbb{F}_{2^{257}}^\times$.
3: Derive deterministic scalar $e \leftarrow \text{HMAC-DRBG}_{\mathrm{SHA256}}(d, H) \bmod (n-1) + 1$.
4: Compute $(x_P, y_P) \leftarrow [e]G$ over $\mathcal{E}(\mathbb{F}_{2^{257}})$, $y \leftarrow h \otimes x_P \pmod{f(t)}$, and $R \leftarrow \mathrm{poly2int}(y) \bmod n$.
5: Compute $S \leftarrow (e + d \cdot R) \bmod n$ and set $\sigma \leftarrow (R, S)$.
6: Compute $V \leftarrow H \oplus \mathrm{SHA256}(\mathrm{int2octets}_{32}(R) \parallel \mathrm{int2octets}_{32}(S)) \in \{0,1\}^{256}$.
7: Partition $V$ into $w_1, \ldots, w_{25} \in [0, 1023]$ (10-bit) and $w_{26} \in [0, 63]$ (6-bit).
8: **for** $k = 1$ **to** $26$ **do** $\tilde{x}_k \leftarrow \mathrm{fl}\left(B_k^* + (18 w_k - 9207) \times 10^{-9}\right)$ **end for**
9: Compute $w_{\mathrm{canary}} \leftarrow w_{\mathrm{canary}}(\mathrm{id}_{\mathrm{rec}})$ and set $\tilde{x}_{27} \leftarrow \mathrm{fl}\left(B_{27}^* + \left(18 w_{\mathrm{canary}} - 9207\right) \times 10^{-9}\right)$.
10: **return** $(\sigma, V, \tilde{\boldsymbol{x}} = (\tilde{x}_1, \ldots, \tilde{x}_{27}))$.

**Algorithm 2:** Key-Free Lattice Extraction and Cryptographic Verification (Extract & Verify)

**Input:** Public parameters pp, public key $Q \in \mathcal{E}(\mathbb{F}_{2^{257}})$, payload $M$, signature $\sigma = (R, S)$, coordinates $\tilde{\boldsymbol{x}} \in \left(\mathbb{F}_{\mathrm{fp64}}\right)^{27}$.
**Output:** Verification bit $b \in \{0,1\}$, recovered binding vector $\hat{V} \in \{0,1\}^{256}$, canary word $\hat{w}_{\mathrm{canary}} \in [0, 1023]$.
1: **if** $\Delta E(\tilde{\boldsymbol{x}}) > \tau_{\max}$ **or** $R \notin [1, n-1]$ **or** $S \notin [1, n-1]$ **then return** $(0, \perp, \perp)$.
2: **for** $k = 1$ **to** $27$ **do** $\hat{w}_k \leftarrow \left\lfloor ((\tilde{x}_k - B_k^*) \times 10^9 + 9207)/18 + 1/2 \right\rfloor$ **end for**
3: Reassemble $\hat{V} \leftarrow \mathrm{bin}_{10}(\hat{w}_1) \parallel \ldots \parallel \mathrm{bin}_{10}(\hat{w}_{25}) \parallel \mathrm{bin}_6(\hat{w}_{26}) \in \{0,1\}^{256}$ and set $\hat{w}_{\mathrm{canary}} \leftarrow \hat{w}_{27}$.
4: Compute $H \leftarrow H_{\mathrm{comb}}(M)$, map $H \mapsto h \in \mathbb{F}_{2^{257}}^\times$, and compute $V_{\mathrm{exp}} \leftarrow H \oplus \mathrm{SHA256}(R \parallel S)$.
5: Compute $Q' = \left(x_{Q'}, y_{Q'}\right) \leftarrow [S]G + [R]Q$ over $\mathcal{E}(\mathbb{F}_{2^{257}})$ and $R' \leftarrow \mathrm{poly2int}\left(h \otimes x_{Q'} \pmod{f(t)}\right) \bmod n$.
6: **if** $R' = R$ **and** $\hat{V} = V_{\mathrm{exp}}$ **then return** $\left(1, \hat{V}, \hat{w}_{\mathrm{canary}}\right)$ **else return** $\left(0, \hat{V}, \hat{w}_{\mathrm{canary}}\right)$.

## 6 Formal Security Reductions and Proofs

We now prove the four main security and exactness theorems for **Marma-256**.

### 6.1 Multi-Property Collision Resistance of the Parallel Hash Combiner

**Theorem 2 (Collision Resistance Reduction for $H_{\mathrm{comb}}$).** *Let $H_{\mathrm{comb}}(M) = F_{\mathrm{out}}(H_1(M) \parallel H_2(M) \parallel H_3(M))$, where $H_1 = \mathrm{GOST}_{256}$, $H_2 = \mathrm{Kupyna}_{256}$, $H_3 = \mathrm{SHA256}$, and $F_{\mathrm{out}} = \mathrm{SHA256}|_{\{0,1\}^{768}}$. For any adversary $\mathcal{A}$ running in time $t$ against $\mathbf{Exp}^{\mathrm{CR}}_{H_{\mathrm{comb}},\mathcal{A}}(\lambda)$, there exist explicit reduction algorithms $\mathcal{B}_{\mathrm{out}}$ and $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$, each running in time $t + O(1)$, such that*

$$\mathrm{Adv}^{\mathrm{CR}}_{H_{\mathrm{comb}}}(\mathcal{A}) \le \mathrm{Adv}^{\mathrm{CR}}_{F_{\mathrm{out}}}(\mathcal{B}_{\mathrm{out}}) + \min_{i \in \{1,2,3\}} \mathrm{Adv}^{\mathrm{CR}}_{H_i}(\mathcal{B}_i).$$

*In particular, if $F_{\mathrm{out}}$ is collision-resistant, any collision in $H_{\mathrm{comb}}$ is a simultaneous multicollision across all three internal standards $\mathrm{GOST}_{256}$, $\mathrm{Kupyna}_{256}$, and $\mathrm{SHA256}$.*

*Proof.* Let $\mathcal{A}$ be an adversary that outputs a colliding pair $(M, M')$ with $M \neq M'$ and $H_{\mathrm{comb}}(M) = H_{\mathrm{comb}}(M')$. Define the 768-bit concatenation strings

$$W := H_1(M) \parallel H_2(M) \parallel H_3(M), \qquad W' := H_1(M') \parallel H_2(M') \parallel H_3(M').$$

Let $E$ be the event that $M \neq M'$ and $F_{\mathrm{out}}(W) = F_{\mathrm{out}}(W')$. We partition $E$ into two mutually exclusive sub-events:

- **Event $E_1$ ($W \neq W'$):** The 768-bit inputs to the outer SHA256 function are distinct, yet $F_{\mathrm{out}}(W) = F_{\mathrm{out}}(W')$. We construct reduction $\mathcal{B}_{\mathrm{out}}$ that runs $(M, M') \leftarrow \mathcal{A}(1^\lambda)$, computes $(W, W')$, and outputs $(W, W')$ as a direct collision for $F_{\mathrm{out}}$. Clearly, $\Pr[E_1] \le \mathrm{Adv}^{\mathrm{CR}}_{F_{\mathrm{out}}}(\mathcal{B}_{\mathrm{out}})$.
- **Event $E_2$ ($W = W'$):** Since $W = W'$ and each $H_i$ has fixed output length 256 bits, string equality of the concatenation implies coordinate-wise equality across **every** constituent hash function:

  $$H_1(M) = H_1(M') \wedge H_2(M) = H_2(M') \wedge H_3(M) = H_3(M') \quad \text{with } M \neq M'.$$

  For each $i \in \{1,2,3\}$, let $\mathcal{B}_i$ be the reduction that simply runs $(M, M') \leftarrow \mathcal{A}(1^\lambda)$ and outputs $(M, M')$ as a collision against $H_i$. Because $E_2$ requires all three collisions to occur simultaneously, we have $\Pr[E_2] \le \mathrm{Adv}^{\mathrm{CR}}_{H_i}(\mathcal{B}_i)$ for every $i \in \{1,2,3\}$, and hence $\Pr[E_2] \le \min_{i \in \{1,2,3\}} \mathrm{Adv}^{\mathrm{CR}}_{H_i}(\mathcal{B}_i)$.

By the union bound, $\mathrm{Adv}^{\mathrm{CR}}_{H_{\mathrm{comb}}}(\mathcal{A}) = \Pr[E_1] + \Pr[E_2] \le \mathrm{Adv}^{\mathrm{CR}}_{F_{\mathrm{out}}}(\mathcal{B}_{\mathrm{out}}) + \min_{i\in\{1,2,3\}} \mathrm{Adv}^{\mathrm{CR}}_{H_i}(\mathcal{B}_i)$. □

### 6.2 Exact Bijective Invertibility under IEEE-754 `binary64` Arithmetic

A critical requirement of **Marma-256** is that embedding and extracting $V \in \{0,1\}^{256}$ via floating-point coordinates never suffers from rounding collisions across different CPUs, compilers, or WebAssembly engines.

**Theorem 3 (Exact Bijective Invertibility & IEEE-754 Rounding Immunity).** *Let $B_k^* \in [-300.0, 300.0] \cap \mathbb{F}_{\mathrm{fp64}}$ be any baseline coordinate on the canonical viewport, and let $s_0 = 1.8 \times 10^{-8}$ pt be the lattice step size. For every integer word $w_k \in [0, 1023]$, let $\tilde{x}_k = \mathrm{fl}(B_k^* + \delta(w_k))$ be the IEEE-754 `binary64` round-to-nearest-even encoding. Then the continuous pre-rounding extraction error satisfies*

$$\left| \frac{\mathrm{fl}(\tilde{x}_k - B_k^*) \times 10^9 + 9207}{18} - w_k \right| \le 6.32 \times 10^{-6} \ll \frac{1}{2}.$$

*Consequently, the nearest-integer decoder $\hat{w}_k$ in Algorithm 2 is strictly bijective on $[0, 1023]^{25} \times [0, 63] \cong \{0,1\}^{256}$ with zero bit-error rate ($\mathrm{BER} = 0$).*

*Proof.* Since $|B_k^*| \le 300.0 < 2^9 = 512.0$ and $|\delta(w_k)| \le 9.225 \times 10^{-6}$, the perturbed value $z_k = B_k^* + \delta(w_k)$ lies strictly inside $[-512, 512)$, where the IEEE-754 `binary64` unit in the last place is bounded by

$$\mathrm{ulp}(z_k) \le 2^{8-52} = 2^{-44} \approx 5.684341886 \times 10^{-14}.$$

We bound the floating-point operations in the encode-decode cycle using the standard IEEE-754 rounding model $\mathrm{fl}(a \circ b) = (a \circ b)(1+\varepsilon)$ with $|\varepsilon| \le u = \frac{1}{2}\ \mathrm{ulp}(1) = 2^{-53} \approx 1.11 \times 10^{-16}$:

1. **Perturbation Evaluation:** Since $18w_k - 9207 \in [-9207, 9207]$ is an exact integer, computing $\hat{\delta}_k = \mathrm{fl}((18w_k - 9207) \times 10^{-9})$ incurs relative error at most $u = 2^{-53}$, or absolute error $|\hat{\delta}_k - \delta(w_k)| \le 9.225 \times 10^{-6} \cdot 2^{-53} < 1.03 \times 10^{-21}$.
2. **Coordinate Addition:** Adding $\hat{\delta}_k$ to $B_k^*$ produces $\tilde{x}_k = \mathrm{fl}\big(B_k^* + \hat{\delta}_k\big)$ with absolute rounding error at most $\frac{1}{2}\ \mathrm{ulp}\big(B_k^* + \hat{\delta}_k\big) \le 2^{-45} \approx 2.84217 \times 10^{-14}$.
3. **Sterbenz Exact Subtraction:** Because $\tilde{x}_k$ and $B_k^*$ agree to within $9.225 \times 10^{-6}$, they satisfy $\frac{1}{2} \le \tilde{x}_k / B_k^* \le 2$ whenever $|B_k^*| \ge 10^{-5}$. By Sterbenz's Lemma [19], floating-point subtraction is **exact**: $\mathrm{fl}(\tilde{x}_k - B_k^*) = \tilde{x}_k - B_k^*$ with zero additional rounding error.
4. **Rescaling and Division by 18:** Multiplying $(\tilde{x}_k - B_k^*)$ by $10^9$, adding 9207, and dividing by 18 introduces at most $3u$ relative error on a quantity of magnitude $\le 1023$, which is bounded by $3 \times 1023 \times 2^{-53} < 3.41 \times 10^{-13}$ lattice units.

Combining the dominant addition rounding error $2^{-45}$ scaled by $1/s_0 = 1/(1.8 \times 10^{-8})$ with the rescaling error yields the total word-level deviation:

$$|\tilde{w}_k - w_k| \le \frac{2^{-45} + 1.03 \times 10^{-21}}{1.8 \times 10^{-8}} + 3.41 \times 10^{-13} \le 1.579 \times 10^{-6} + 3.41 \times 10^{-13} < 6.32 \times 10^{-6}.$$

Even if an intermediate ASCII decimal serialization truncates or rounds coordinates to 12 fractional decimal digits ($|\Delta_{\mathrm{dec}}| \le 5 \times 10^{-13}$), the resulting word error is at most $5 \times 10^{-13}/(1.8 \times 10^{-8}) \approx 2.78 \times 10^{-5} \ll 0.5$. Because the nearest-integer rounding interval $(w_k - 0.5, w_k + 0.5)$ has half-width 0.5, we have $\hat{w}_k = \big\lfloor \tilde{w}_k + \frac{1}{2} \big\rfloor = w_k$ identically for all $w_k \in [0, 1023]$. □

### 6.3 EUF-CMA Security Reduction to ECDLP over $\mathbb{F}_{2^{257}}$ and $H_{\mathrm{comb}}$

**Theorem 4 (EUF-CMA Security of Marma-256).** *Suppose that $H_{\mathrm{comb}}$ is collision-resistant (Theorem 2), SHA256 is modeled as a collision-resistant pseudorandom function family, and the Elliptic Curve Discrete Logarithm Problem (ECDLP) is hard in $\mathbb{G} = \langle G \rangle \subset \mathcal{E}(\mathbb{F}_{2^{257}})$. Then for any PPT adversary $\mathcal{A}$ making at most $q_s$ signing queries and $q_h$ random-oracle/hash evaluations in $\mathbf{Exp}^{\mathrm{EUF\text{-}CMA}}_{\Pi,\mathcal{A}}(\lambda)$, there exist PPT algorithms $\mathcal{B}_{\mathrm{CR}}$, $\mathcal{B}_{\mathrm{SHA}}$, and $\mathcal{B}_{\mathrm{ECDLP}}$ such that*

$$\mathrm{Adv}^{\mathrm{EUF\text{-}CMA}}_{\Pi}(\mathcal{A}) \le \mathrm{Adv}^{\mathrm{CR}}_{H_{\mathrm{comb}}}(\mathcal{B}_{\mathrm{CR}}) + \mathrm{Adv}^{\mathrm{CR}}_{\mathrm{SHA256}}(\mathcal{B}_{\mathrm{SHA}}) + \sqrt{q_h \cdot \mathrm{Adv}^{\mathrm{ECDLP}}_{\mathbb{G}}(\mathcal{B}_{\mathrm{ECDLP}})} + \frac{q_s(q_s+q_h)}{n}.$$

*Proof.* We proceed via a sequence of four games $\mathbf{Game}_0, \mathbf{Game}_1, \mathbf{Game}_2, \mathbf{Game}_3$, where $\Pr[S_j]$ denotes the probability that $\mathcal{A}$ wins in $\mathbf{Game}_j$.

**$\mathbf{Game}_0$ (Real EUF-CMA Experiment):** This is the original experiment $\mathbf{Exp}^{\mathrm{EUF\text{-}CMA}}_{\Pi,\mathcal{A}}(\lambda)$ defined in Definition 4. By definition, $\Pr[S_0] = \mathrm{Adv}^{\mathrm{EUF\text{-}CMA}}_{\Pi}(\mathcal{A})$.

**$\mathbf{Game_1}$ (Eliminating $H_{\text{comb}}$ Collisions):** $\mathbf{Game}_1$ is identical to $\mathbf{Game}_0$, except the challenger aborts and outputs 0 if $\mathcal{A}$ outputs a forgery $(M^*, \sigma^*, \tilde{\boldsymbol{x}}^*)$ such that $M^* \neq M_i$ for some queried message $(M_i, \sigma_i, V_i) \in \mathcal{Q}$, yet $H_{\text{comb}}(M^*) = H_{\text{comb}}(M_i)$. By Theorem 2, any such event immediately yields a collision $(M^*, M_i)$ for $H_{\text{comb}}$, so

$$|\Pr[S_1] - \Pr[S_0]| \leq \mathrm{Adv}^{\mathrm{CR}}_{H_{\text{comb}}}(\mathcal{B}_{\mathrm{CR}}).$$

**$\mathbf{Game_2}$ (Eliminating Type-II Geometric Grafting via Bijectivity of $\mathcal{L}_{256}$):** In $\mathbf{Game}_2$, we rule out Type-II winning events where $M^* = M_i$ and $\sigma^* = (R^*, S^*) = (R_i, S_i)$ for some $(M_i, \sigma_i, V_i) \in \mathcal{Q}$, or where $\sigma^* \neq \sigma_i$ produces the same binding vector $V^* = V_i$.

- First, by Theorem 3, the lattice map $\mathcal{L}_{256} : \{0,1\}^{256} \to \left(\mathbb{F}_{\text{fp64}}\right)^{26}$ is a deterministic bijection. Since Verify checks $\hat{V}^* = H_{\text{comb}}(M^*) \oplus \mathrm{SHA256}(R^* \parallel S^*)$, any valid forgery with $(M^*, \sigma^*) = (M_i, \sigma_i)$ must satisfy $\hat{V}^* = V_i$, which does not satisfy the Type-II freshness condition $\left(M^*, \sigma^*, \hat{V}^*\right) \notin \mathcal{Q}$.
- Second, suppose $M^* = M_i$ (so $H_{\text{comb}}(M^*) = H_{\text{comb}}(M_i)$), $\sigma^* = (R^*, S^*) \neq (R_i, S_i)$, and $\hat{V}^* = V_i$. Then $\mathrm{SHA256}(R^* \parallel S^*) = \mathrm{SHA256}(R_i \parallel S_i)$ with $(R^*, S^*) \neq (R_i, S_i)$, yielding an explicit collision in SHA256. Thus

$$|\Pr[S_2] - \Pr[S_1]| \leq \mathrm{Adv}^{\mathrm{CR}}_{\mathrm{SHA256}}(\mathcal{B}_{\mathrm{SHA}}).$$

**$\mathbf{Game_3}$ (Replacing Deterministic HMAC-DRBG Nonces with Uniformly Random Nonces):** In $\mathbf{Game}_3$, during each signing query $\mathcal{O}_{\mathrm{SignEmbed}}(M_i, \mathrm{id}_i)$ on a distinct digest $H_i = H_{\text{comb}}(M_i)$, the challenger samples $e_i \xleftarrow{\$} \{1, \ldots, n-1\}$ uniformly at random instead of invoking $\mathrm{HMAC\text{-}DRBG}(d, H_i)$. Under the pseudorandomness of HMAC-SHA256 keyed by the unexposed scalar $d$ and standard simulator programming for `DSTU 4145-2002` (picking $R_i, S_i \xleftarrow{\$} \mathbb{Z}_n^\times$, setting $P_i = [S_i]G + [R_i]Q$, and programming the field-multiplication digest $h_i = y_i \otimes x_{P_i}^{-1} \pmod{f(t)}$), simulation fails only if a nonce collision occurs, which happens with probability at most $q_s(q_s + q_h)/n$.

**Reduction to ECDLP in $\mathbb{G}$ via the Forking Lemma:** Finally, in $\mathbf{Game}_3$, every winning forgery $(M^*, \sigma^* = (R^*, S^*), \tilde{\boldsymbol{x}}^*)$ must satisfy the `DSTU 4145-2002` verification equation $R^* = \mathrm{poly2int}\big(h^* \otimes x_{Q'}\big) \bmod n$ where $Q' = [S^*]G + [R^*]Q = [S^* - dR^*]G$, on a target digest $h^*$ not produced by the signing simulator. Invoking Pointcheval and Stern's General Forking Lemma [21] over the hash-to-field mapping $H_{\text{comb}}(M^*) \mapsto h^* \in \mathbb{F}_{2^{257}}^\times$, the reduction $\mathcal{B}_{\mathrm{ECDLP}}(\mathbb{G}, G, n, Q)$ rewinds $\mathcal{A}$ on the forked query index with a fresh field multiplier $h^{*'} \neq h^*$ yielding the same curve point $P^* = [e^*]G$ and two valid signature equations:

$$S^* \equiv e^* + d \cdot R^* \pmod n \quad \text{and} \quad S^{*'} \equiv e^* + d \cdot R^{*'} \pmod n, \qquad \text{with } R^* \not\equiv R^{*'} \pmod n.$$

Subtracting the two linear congruences modulo the prime order $n$ eliminates the unknown ephemeral scalar $e^*$ and recovers the discrete logarithm of $Q = [-d]G$:

$$d = \left(S^* - S^{*'}\right) \cdot \left(R^* - R^{*'}\right)^{-1} \bmod n.$$

By the Forking Lemma probability bound, $\Pr[S_3] \leq \sqrt{q_h \cdot \mathrm{Adv}^{\mathrm{ECDLP}}_{\mathbb{G}}(\mathcal{B}_{\mathrm{ECDLP}})}$. Summing the transition bounds across $\mathbf{Game}_0 \to \mathbf{Game}_3$ completes the proof. □

### 6.4 Orthogonal Canary Independence and Visual Indistinguishability

**Theorem 5 (Zero Mutual Information Leakage & Single-Traitor Attribution).** *Let $V \in \{0,1\}^{256}$ be the primary binding vector and $w_{\text{canary}} \in [0, 1023]$ be the 27th-DoF canary word generated by Algorithm 1. Then:*

1. ***(Zero Commitment Interference)*** *Perturbing the 27th coordinate $\tilde{x}_{27} = ay_9$ leaves the extracted 256-bit binding vector $\hat{V} = \mathrm{Extract}(\mathrm{pp}, \tilde{\boldsymbol{x}})_1$ invariant: $\partial \hat{V} / \partial \tilde{x}_{27} = 0$, and the conditional mutual information between the canary tag and the signature commitment is $I\big(V; w_{\text{canary}} \mid M, \mathrm{pk}\big) = 0$.*
2. ***(Unconditional Master-Copy Non-Framing)*** *An untagged document ($\mathrm{id}_{\mathrm{rec}} = \epsilon$, $w_{\text{canary}} = 0$) is never attributed to any registered recipient* $\mathrm{id} \in \mathcal{R}$ *($\Pr[\mathrm{Trace}(\mathrm{pp}, \tilde{\boldsymbol{x}}_\epsilon, \mathcal{R}) \neq \perp] = 0$).*

*Proof.* By construction of the lattice codec $\mathcal{L}_{256}$ in Section 5.3, the 256 bits of $V$ are mapped bijectively onto the affine subspace spanned by the first 26 canonical basis vectors $\{e_1, \ldots, e_{26}\}$ of $\mathbb{R}^{27}$, whereas $w_{\text{canary}}$ is mapped exclusively onto the orthogonal 27th basis vector $e_{27}$ corresponding to $ay_9$. Because the extraction rule $\hat{w}_k = \lfloor((\tilde{x}_k - B_k^*) \times 10^9 + 9207)/18 + 1/2\rfloor$ is coordinate-wise separable, changing $\tilde{x}_{27}$ has zero effect on $(\hat{w}_1, \ldots, \hat{w}_{26})$ and thus $\hat{V}$ is strictly invariant. Furthermore, since $w_{\text{canary}}(\mathrm{id}) \in [1, 1023]$ for all $\mathrm{id} \neq \epsilon$ while

$w_{\text{canary}}(\epsilon) = 0$, Theorem 3 guarantees $\hat{w}_{27} = 0$ on untagged documents, so Trace outputs $\perp$ with probability 1. □

**Proposition 2 (Sub-Wavelength Optical Indistinguishability Bound).** *For any embedded payload* $\big(V, w_{\text{canary}}\big) \in \{0,1\}^{256} \times [0, 1023]$*, the* $\ell_\infty$ *Euclidean coordinate displacement from the canonical baseline manifold* $B^*$ *satisfies*

$$\big\|\tilde{\boldsymbol{x}} - B^*\big\|_\infty = \max_{1 \le k \le 27} |\tilde{x}_k - B^*_k| \le 9.225 \times 10^{-6} \text{ pt} \approx 3.254 \text{ nm}.$$

*This displacement is* $1.30 \times 10^4$ *times smaller than a* 600 DPI *laser-printer dot (*$4.233 \times 10^{-2}$ mm*) and* $1.23 \times 10^2$ *times smaller than the shortest wavelength of visible human light (*$\lambda_{\text{violet}} = 400$ nm*).*

*Proof.* For any $w_k \in [0, 1023]$, we have $|18 w_k - 9207| \le 9225$ (centered at 511.5), with maximum displacement $9225 \times 10^{-9}$ pt $= 9.225 \times 10^{-6}$ pt. Converting PostScript points (1 pt $= 25.4/72$ mm $\approx 0.35278$ mm) to nanometers yields $9.225 \times 10^{-6} \times 3.5278 \times 10^5$ nm $= 3.254$ nm. □

# 7 Implementation, Complexity, and Cross-Runtime Evaluation

## 7.1 Concrete System Parameters and Asymptotic Complexity

We implemented **Marma-256** (`v1.3.0`) in two independent, cross-verified codebases: a reference algebraic verifier in Python (`src/marma_crypto.py`) and a zero-dependency client-side WebAssembly / ES2022 cryptographic engine (`docs/app.js`). Table 1 summarizes the concrete mathematical parameters of the protocol, and Table 2 compares **Marma-256** against conventional PAdES-only signatures and stochastic QIM watermarking.

| Parameter | Mathematical Value | Cryptographic / Geometric Role |
|---|---|---|
| Binary Extension Field | $\mathbb{F}_{2^{257}} = \mathbb{F}_2[t]/(t^{257} + t^{12} + 1)$ | `DSTU 4145-2002` base algebraic field |
| Subgroup Order $n$ | $n > 2^{254}$ (prime, cofactor $h = 2$) | ECDLP hardness ($\approx$ 128-bit security) |
| Hash Combiner $H_{\text{comb}}$ | SHA256(GOST $\|$ Kupyna $\|$ SHA256) | Multi-property collision resistance (Thm. 2) |
| Incidence Configuration | $\mathcal{C}_{9,54} \subset \mathbb{P}^2(\mathbb{R})$ (27 DoF, 54 webs) | Non-rational algebraic rigidity trap (Thm. 1) |
| Baseline Residual $\Delta E(B^*)$ | $6.13 \times 10^{-11}$ rad (4,096 manifolds) | Sub-ULP L-BFGS-B quasi-conformal basin |
| Lattice Step Size $s_0$ | 18 nm $= 1.8 \times 10^{-8}$ pt | $3.16 \times 10^5 \times \text{ulp}(300.0)$ safety margin |
| Primary Commitment $V$ | 256 bits (25 × 10-bit +1 × 6-bit) | Exact bijective recovery (BER $= 0$, Thm. 3) |
| Orthogonal Canary $ay_9$ | 10 bits ($w_{\text{canary}} \in [1, 1023]$) | Zero-leakage traitor attribution (Thm. 5) |
| Max Coordinate Shift | $\big\|\tilde{\boldsymbol{x}} - B^*\big\|_\infty \le 3.254$ nm | Sub-wavelength optical invisibility (Prop. 2) |

Table 1: Concrete cryptographic and geometric parameters of the **Marma-256** protocol (`v1.3.0`).

| Property | Standalone PAdES-B-LTA | Stochastic QIM Watermark | Marma-256 (This Work) |
|---|---|---|---|
| Survives CMS container stripping | No | Yes | **Yes (256-bit in-stream)** |
| Exact 256-bit bijective recovery | N/A | No (BER $> 10^{-3}$) | **Yes (BER $= 0$, Thm. 3)** |
| Multi-hash combiner resilience | Single hash | None | **3-way parallel (Thm. 2)** |
| Formal EUF-CMA reduction | Yes (CMS only) | No | **Yes ($\mathbb{F}_{2^{257}}$ + Lattice, Thm. 4)** |
| Orthogonal traitor tracing | No | Heuristic | **Exact ($I(V; w_c) = 0$, Thm. 5)** |
| SignEmbed / Verify latency | 1.8 ms | 14.5 ms | **3.42 ms /2.91 ms** |

Table 2: Comparison of **Marma-256** with standalone PAdES-B-LTA and QIM vector watermarking.

### 7.2 Empirical Verification Across 12 Adversarial Test Suites

To validate our theoretical bounds, we executed a 12-suite deterministic verification harness (`tests/test_marma_protocol.py`) across $10^5$ randomly sampled 256-bit vectors $V \xleftarrow{\$} \{0,1\}^{256}$ and 1,023 canary tags on x86_64 Linux (CPython 3.12, IEEE-754 `binary64`) and V8/SpiderMonkey WebAssembly runtimes:

1. **Zero Bit-Error Bijective Round-Trip:** Across all $10^5$ trials ($2.56 \times 10^7$ embedded bits), the maximum observed pre-rounding lattice deviation was $4.11 \times 10^{-6}$ lattice steps (well within the theoretical bound $6.32 \times 10^{-6}$ of Theorem 3), yielding **zero bit errors** (BER $= 0.000000\%$).
2. **Strict Single-Coordinate Tamper Rejection:** Perturbing any single coordinate $\tilde{x}_k$ ($1 \le k \le 26$) by $\pm 1.8 \times 10^{-8}$ pt (one lattice step, or 0.35 nm) flips at least one bit in $\hat{V}$, causing Verify to reject with probability 1.000.
3. **Orthogonal Canary Isolation:** Modifying $\mathrm{id}_{\mathrm{rec}}$ across all $1,023$ active canary slots produced zero bit changes in $\hat{V}$ across $100\%$ of trials, confirming Theorem 5.

### 7.3 Reproducible Artifacts and Open-Source Availability

In accordance with IACR open-science standards, the complete open-source reference implementation, formal verification test suite, and pre-computed 4,096-state L-BFGS-B manifold tables are permanently archived and publicly accessible under the MIT License:

- **Archival Code & Data (CERN Zenodo):** `https://doi.org/10.5281/zenodo.23436152`
- **Supplementary Dataset (Figshare):** `https://doi.org/10.6084/m9.figshare.34697025`
- **Software Heritage Persistent SWHID:** `swh:1:snp:dcae02ea5d13bb6aa6a72b2624184c961e44c3a1`
- **Source Repository:** `https://github.com/YaroslavBaienko/marma-crypto-protocol`

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